

FitBuddy – AI Fitness Plan Generator

Project Blueprint, Technical Architecture & Source Code

SYSTEM TECHNICAL SPECIFICATION REPORT

1. Project Description

Creating customized fitness and nutrition routines tailored to an individual's goals, dietary restrictions, biometric profile, and fitness levels is complex and time-consuming. Users often struggle to build safe, effective workout plans and calculated meal targets that align with their available equipment and busy schedules.

FitBuddy addresses this challenge through a GenAI-powered fitness generation platform that delivers personalized, goal-aware workout schedules, caloric/macronutrient plans, and real-time exercise posture tips. Using **Google Gemini 2.5 Flash** for multimodal reasoning, the platform analyzes user input parameters (and optional body composition or workout space photos) to generate tailored routines with step-by-step guidance.

FitBuddy transforms workout planning into an intelligent, efficient process through smart calorie calculation, adaptive split generation, and multi-format export capabilities.

2. Use Case Scenarios

Scenario 1: Home Workout Optimization with Equipment Constraints

User: A busy professional with only dumbbells and resistance bands wants a 4-day muscle-building split.

Flow: The user inputs biometrics, equipment constraints, and weekly commitment. FitBuddy's Gemini backend generates a structured 4-day Upper/Lower split, replacing barbell movements with dumbbell/band alternatives, calculating maintenance and surplus calories, and generating a daily meal split with macro targets.

Scenario 2: Multimodal Visual Space & Posture Analysis

User: A beginner uploads a photo of their home workout space to check adequacy and a photo of their squat stance for alignment feedback.

Flow: Gemini Vision analyzes the space, confirms safety clearance, detects stance flaws (e.g., knee cave), and updates the plan with dynamic warm-ups and corrective exercise cues tailored to posture.

Scenario 3: Custom Dietary Restrictions & Caloric Planning

User: A vegan user targeting weight loss with high-protein preferences.

Flow: The engine calculates total daily energy expenditure (TDEE) and target deficit calories, structuring a vegan meal plan complete with micronutrient advice, protein source pairings, and automatic smart formatting.

3. Component-Wise Architecture

Component Module	Technical Function & Responsibility
Frontend UI	Streamlit-based interface with multi-page input, photo upload, plan customization, and interactive metric displays. Includes DOCX/PDF exports.
Biometric & TDEE Calculator	Calculates BMR, TDEE, and target intake using Mifflin-St Jeor / Katch-McArdle equations to establish foundational caloric baselines.
Workout Plan Generator	Creates structured weekly routines (Push/Pull/Legs, Upper/Lower, Full Body) with targeted sets, reps, REST times, and progressions using Gemini 2.5 Flash.
Nutrition & Macro Engine	Generates meal options aligned with calorie targets, dietary constraints (Vegan, Keto, Paleo, Halal), and macro split ratios.
Gemini Vision Analyzer	Processes user-submitted space or posture images via Gemini 2.5 Flash to provide visual exercise safety and stance recommendations.
Exporter Module	Exports generated fitness and nutrition schedules into styled .docx files.

4. System Architecture Diagram



5. Prerequisites & Workflow Milestones

Prerequisites: Python 3.10+, Streamlit Framework, Google AI Studio API Key (for Gemini 2.5 Flash), and standard libraries (``google-genai``, ``python-docx``, ``pillow``, ``pydantic``).

- **Milestone 1: Environment Setup & Configuration** — Directory structure setup, virtual environment, dependency management, and API key verification.
- **Milestone 2: Core Services Development** — TDEE math engine, Workout Routine Generator, Nutrition Engine, and Multimodal Vision logic.
- **Milestone 3: Backend & Session Integration** — Streamlit session state management, prompt construction, and error fallback handlers.
- **Milestone 4: Streamlit UI Implementation** — Multi-tab controls, metric badges, progress bars, and document download interfaces.
- **Milestone 5: Testing & Performance Optimization** — Caching model instances via `@st.cache_resource`` for high performance.

6. Implementation Source Code

```
requirements.txt    config/settings.py    services/gemini_service.py
services/workout_service.py    services/nutrition_service.py    utils/export_utils.py
app.py
```

```
# Dependencies for FitBuddy Application
streamlit>=1.31.0
google-genai>=0.1.1
python-docx>=1.1.0
pillow>=10.0.0
python-dotenv>=1.0.0
pydantic>=2.0.0
```

7. Conclusion

FitBuddy modernizes personal fitness and nutrition planning by bridging advanced Generative AI with biometric data and visual analysis. Powered by Google's Gemini 2.5 Flash, the platform delivers highly personalized, adaptive workout routines, precision caloric and macronutrient breakdowns, and real-time exercise form feedback.

By simplifying complex fitness logic into an accessible Streamlit interface with document export capabilities, FitBuddy empowers users to achieve their physical health goals safely

and efficiently.